Testing by betting when the bets are real: an e-value audit of 84 pre-registered market-efficiency hypotheses

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Abstract

E-values are motivated by a gambler betting against a null hypothesis. We report an audit where the gambler is not a metaphor: the null is a bookmaker's posted price, the stake is a stake, and the wealth process *is* the e-process. The subject is a deployed UFC forecaster and a pre-registered search for pockets where it beats the closing line: 84 hypotheses over 1,787 priced bouts, originally analysed with paired log-loss and Benjamini–Hochberg. Re-asked as bets, three things change. (i) The 84 slices are overlapping subsets of one pool, so BH's guarantee is not established; its one favourable discovery survives neither Benjamini–Yekutieli nor e-BH. (ii) A segment the fixed-n analysis reported as *failing to replicate* ($p = 0.30$) is, on the wealth scale, growing at the same rate in the replication window and 100 bouts short of $1/\alpha$: optional continuation converts a refutation into a schedule. (iii) A post-hoc threshold rule, paid for by a mixture martingale, survives the search it actually ran ($e = 31.3$) and dies twice over — once when the direction it could also have searched is priced in (15.6), and again when the bookmaker's margin is charged (8.5). We report the ladder, not the winner.

1 A null hypothesis with money behind it

Testing by betting [Shafer, 2021, Grünwald et al., 2024, Vovk and Wang, 2021, Ramdas et al., 2023] introduces its central object through a gambler whose capital cannot grow in expectation if the null is true. The gambler is usually a device. Here it is the situation itself.

We audit VERTEX, a deployed UFC bout-outcome model, against a betting exchange's closing line. The substrate is a walk-forward out-of-fold pool: $4{,}141$ bouts scored between 2016 and 2026 by models refit at 32 quarterly origins, $1{,}787$ of them across 225 events carrying an archived closing price. Two properties no simulation supplies:

The null is adversarial and external. H_0 is not an assumption of convenience — it is a number a counterparty posted with capital at risk, and a bettor who rejects it collects. Under H_0 the de-vigged closing probability q_t — the two closing decimals stripped of their overround by the power method, solving $q_a^k + q_b^k = 1$ per bout — is the true conditional probability of the outcome $y_t \in \{0, 1\}$. **Predictability is enforced by the calendar.** The model probability p_t comes from a fit that saw only bouts strictly before its origin, and q_t is posted before the fight; neither has to be argued into being predictable.

The prior analysis of this pool [Prigodskii, 2026] declared 86 candidate segments, each with a stated mechanism for why a good market would be wrong in that exact place, in a registry[1] written before

[1] Pre-registration here means author-controlled version history, not third-party registration: the registry is one inspectable source file whose commit (`8f9fcbd`) precedes the first scored artifact (`6b5fbd9`) in a public repository. That fixes the

any of them was scored. Two pairs are the same filter proposed by different lenses, so the pre-registered family is 84; three further segments, added once results were visible, are marked post-hoc there and excluded from multiplicity accounting here, and $84 + 3$ is the 87 its title counts. It measured each with a composition-adjusted paired log-loss contrast, cluster-robust by event, and charged multiplicity with BH — an instrument whose own power section reports 10–15% against a 3 percentage-point edge, 0–1% *once BH multiplicity is charged*, and ∼89,900 bouts needed. There are 1,787. That report is the reason for this one: the instrument was known to be inadequate before its answers were read.

2 The wealth process

For a bout with de-vigged closing probability q and outcome y, and a predictable stake λ,

$$M \;=\; 1 + \lambda\,(y - q), \qquad \lambda \in \left(-\tfrac{1}{1-q},\, \tfrac{1}{q}\right), \qquad \mathbb{E}_{H_0}[M] = 1. \tag{1}$$

The product over a segment's bouts *in calendar order* is a non-negative test martingale, so its terminal value is an e-value and, by Ville's inequality [Ville, 1939], $\Pr_{H_0}(\sup_t M_t \geq 1/\alpha) \leq \alpha$: it may be stopped at any time. Every segment filter is a function of pre-fight covariates, so membership is $\mathcal{F}_{t-1}$-measurable, and restricting the product to a segment is optional skipping ($\lambda_t = 0$ off the segment), which preserves the martingale property. The growth-rate optimal stake under the alternative "the model's p is the truth" is available in closed form as $\lambda^\star = (p - q)/[q(1 - q)]$, the model's edge deflated by the market's own variance. We report fractional Kelly at $\lambda = \frac{1}{2}\lambda^\star$ and, because that fraction is a free choice, a predictable plug-in [Waudby-Smith and Ramdas, 2024] that learns the multiplier from past bouts only (§7).

The margin is a supermartingale, not an excuse. Write $f = \lambda q$ for the fraction of wealth the stake commits to the backed side and d for the decimal the book offers on it; a fair book would offer $d = 1/q$. Settling at d gives $\mathbb{E}_{H_0}[M] = 1 - f(1 - q\,d) \leq 1$, since $q\,d < 1$ whenever the book charges an overround: the deployable process is a valid e-value, uniformly conservative, short of the statistical one by exactly what the margin costs. On this pool the mean overround is 1.0482 and the shortfall 0.00525 nats per bout, so every e-value below is given at both prices, *fair* and *real* — over a 300-bout segment they differ by a factor of seven.

Validation before use. Under a true null — same prices, same stakes, only the outcome resampled from q — 40,000 replications give $\mathbb{E}[e] = 0.985 \pm 0.018$ per segment, $\Pr(\sup_t e_t \geq 1/\alpha)$ of 0.029/0.067/0.151 at $\alpha = 0.05/0.10/0.20$, none of the 84 above its own α, e-BH rejecting at all in 74 of the runs, and the confidence sequence of §6 covering the truth at *every* t in 97.4% of them.

Where the model actually stands. Betting the model against the book over the whole priced pool gives $e = 3.6 \times 10^{-4}$ at fair odds and 3.1×10^{-8} at real ones. This is not a paper about beating a market, but about what can be honestly claimed inside a search that mostly failed.

3 The multiplicity charge BH was not licensed to make

The 84 slices are overlapping subsets of one pool of 1,787 bouts, scored by one model against one book: a single upset moves dozens of them together. BH's FDR guarantee requires independence or PRDS [Benjamini and Yekutieli, 2001], and neither is established here. Nor is the dependence of the benign kind that would make PRDS plausible without an argument: across the 3,486 segment pairs, membership correlations run from −0.49 to +0.84, half of them negative, and 83 pairs share no bouts at all. The classical repair is Benjamini–Yekutieli, which charges $\sum_{i\leq 84} 1/i = 5.01$. e-BH [Wang and Ramdas, 2022] is valid under *arbitrary* dependence and needs no dependence correction at all: its charge is K/α whatever the correlation structure turns out to be. On the discovery window the lab searched (1,191 bouts, the same 84 hypotheses):

procedure	guarantee holds?	rejects	
BH, paired statistic	not established	4	1 favourable, 3 against
Benjamini–Yekutieli (×5.01)	yes	0	
e-BH, wealth statistic	yes	0	cutoff $e \geq 1680$ (k=1)

hypotheses before they were scored, not before any contact with a pool already months in use — and no timestamp available to us would.

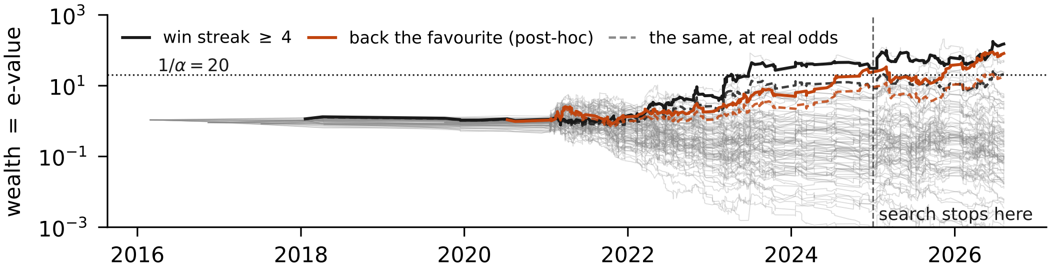

Figure 1: Wealth over calendar time for every pre-registered segment (grey) and the two the analysis singled out (coloured); dashed, the same bets at the book's real prices. At the vertical rule the search stopped and the replication window began: a fixed-n analysis must start a new test to the right of it, a test martingale need not.

The lab's single favourable discovery — a fighter carrying a win streak of four or more, BH $q = 0.016$ — does not survive either procedure whose guarantee actually holds. It would be wrong to read this as e-values buying less. The same segment's e-value on the full pool is 86.4 at fair odds — the median over the pool's five walk-forward seeds — and 12.3 at real ones, and for a *single* pre-specified hypothesis $e \geq 20$ already rejects at $\alpha = 0.05$. What is expensive is not the machinery but the charge for asking 84 questions of a pool this size — the original analysis's own lesson, in a currency where the arithmetic is visible.

4 "Failed to replicate" and "not yet" are different findings

The original analysis froze its discovery window, selected the streak segment, and ran a fresh fixed-n test on 2025–26. It returned $p = 0.30$ and was reported, correctly under its own logic, as a failure to replicate — the only move a fixed-n design has, since the second window buys a second test and nothing carries over.

On the wealth scale the same 110 bouts read differently. Log-growth in the replication window is $+0.0143$ nats per bout against $+0.0174$ in the window that selected it — 82% of the discovery rate, the shrinkage one expects when a slice is chosen on the data that flatters it, but the same sign and order. The e-value accumulated there is 4.81 — real evidence, short of $1/\alpha = 20$. At that rate, crossing needs 210 bouts and 110 have arrived — an extrapolation at a rate estimated on that same window, not a guarantee: the e-process is valid whenever it is stopped, but the date it crosses on is a forecast. The finding is not refuted; it is 100 bouts short, and at about 71 a year that is seventeen months of continuing to multiply rather than choosing a stopping time in advance (Figure 1). At real odds the window pays 2.48, the margin taking 0.0060 of the 0.0143, and the wait is 253 bouts: three and a half years. We report the pre-split wealth (31.4 fair, 8.5 real) but do not claim it: that is the window in which the segment was chosen. The effect is stable across the pool's five walk-forward seeds — fair 56.2 to 150.9, all above $1/\alpha$; real 8.4 to 21.2, where only the most favourable clears it — a spread q-values had no natural place to put.

5 Pricing the freedom a post-hoc rule used

The largest effect in the original analysis is not a segment but a direction: where the model gives the book's favourite *more* than the book does, flat-staking that favourite returned $+11.5\%$ over 281 bets across 163 events — declared post-hoc there, because the cut at $\mathrm{lean} \geq 0.05$ was chosen after seeing the sweep was positive from 0.02 to 0.22 and peaked at 0.13, and a fixed-n framework has no principled penalty for that.

A mixture martingale has one, with weights that do not depend on which cut won [Robbins, 1970, Howard et al., 2021]: each threshold's wealth process is a martingale starting at 1, so any convex combination is too, and the search is paid for by the prior mass not on the winner. The maximum over thresholds, 118.99, is not a valid e-value. The uniform mixture over the 21 cuts is $e = 31.3$, which clears $1/\alpha$. Widening the family to the 21 cuts the search could also have taken in the opposite direction gives 15.6, which does not. Adding the other disagreement statistic available on the same

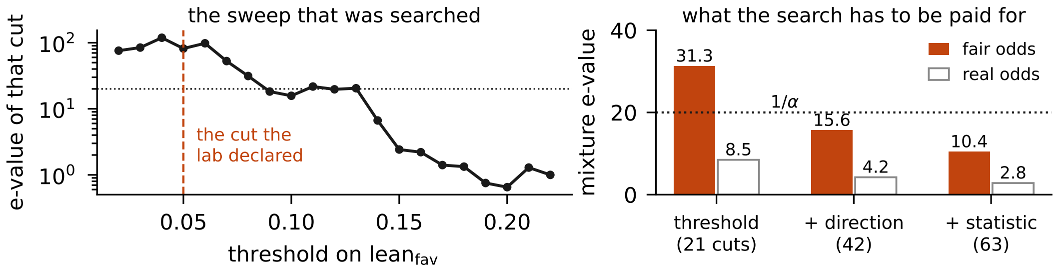

Figure 2: Left: the e-value of each threshold on the model's lean toward the book's favourite; the declared cut at 0.05 was chosen after this shape was visible. Right: a mixture over progressively more of the freedom the search had. At fair odds the finding pays for the threshold but not the direction; at real prices, neither.

frame gives 10.4. Those are fair odds; at the prices the book offered, the same mixtures are 8.5, 4.2 and 2.8 — a bettor paying the margin never clears $1/\alpha$ at all.

We regard the ladder, not any rung, as the result: it turns "this was post-hoc" from a caveat into a quantity. The finding absorbs the threshold search at fair odds, and neither the direction it could have taken nor the margin it would have paid.

6 What the sequential instrument needs

A growth rate makes planning a calculation rather than a verdict. Against a true 3 pp edge simulated on this pool's own prices, the half-Kelly bet grows at 0.00166 nats per bout, so expected log-wealth reaches $\log(1/\alpha)$ in $\sim$1,800 bouts — a 59% chance of having crossed by then, not a power level. Matched at 80% over $20{,}000$ simulated runs, the sequential test needs $2{,}385$ bouts to the paired log-loss test's $5{,}346$; at 5 pp, 864 to $1{,}902$. It is roughly twice as cheap in fights, not because it is more powerful per observation, but because a paired contrast on a slice discards most of what each bout carries.

Anytime-validity is not free. On the directional rule the cluster bootstrap gives a 95% ROI interval of $[+2.2\%, +20.4\%]$, valid at the one n the analysis stopped at — an n chosen after watching the number. The hedged betting confidence sequence [Waudby-Smith and Ramdas, 2024], valid at every n at once, gives $[-5.3\%, +22.9\%]$ and does not exclude zero.

7 Limitations

The stake and the de-vig are both free choices, and neither carries the result. Fractional Kelly at $0.25/0.5/1.0$ gives $e = 26.0/150.9/115.3$ on the streak segment's primary seed and the predictable plug-in, which is not a choice, settles at $c = 0.70$, $e = 63.1$; every variant clears $1/\alpha$. Under Shin's de-vig the rungs are 120.6, 4.36 and $34.0/17.0/11.3$; the proportional split, which flatters a favourite-backing rule, would *raise* the ladder to $56.3/28.2/18.8$, not lower it.

Boosting e-BH [Wang and Ramdas, 2022] would recover power but needs each e-value's null distribution, which a composite martingale null does not supply; weights fixed in advance would be the better route, and were not fixed.

Three contaminations are inherited from the pool and sized there: an age correction shipped on replication-window data; hyperparameters tuned on a window the pool later scores (122 of $1{,}191$ discovery bouts); a rating not fully point-in-time (21.9% of bouts, 0.0011 nats). Removing any moves the numbers unfavourably. And this is one sport, one book, one model: the construction generalises, the numbers do not.

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